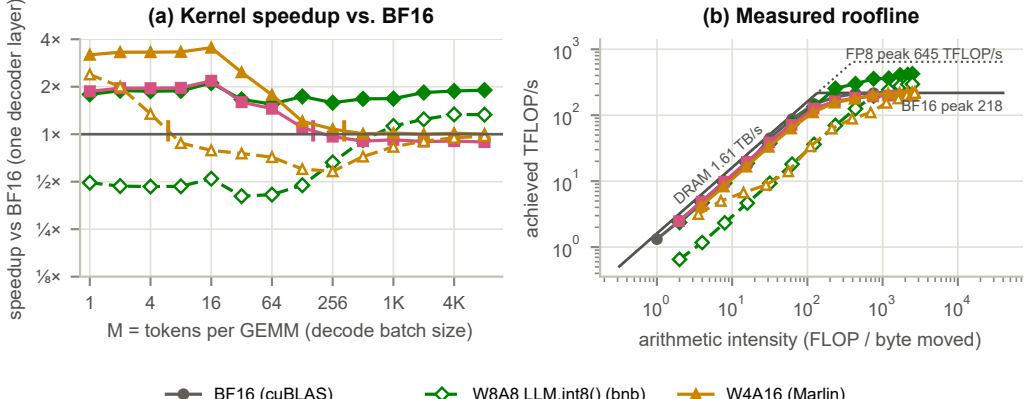

(a) Kernel speedup vs. BF16
speedup vs BF16 (one decoder layer)
4×
2×
1×
½×
¼×
⅛×
1
4
16
64
256
1K
4K
M = tokens per GEMM (decode batch size)
(b) Measured roofline
achieved TFLOP/s
10^3
10^2
10^1
10^0
FP8 peak 645 TFLOP/s
BF16 peak 218
DRAM 1.61 TB/s
10^0
10^1
10^2
10^3
10^4
arithmetic intensity (FLOP / byte moved)
BF16 (cuBLAS)
W8A8 FP8 (CUTLASS)
W8A8 LLM.int8() (bnb)
W8A16 (Marlin)
W4A16 (Marlin)
W4A16 NF4 (bnb)